Assignemet1 Data Mining

1 Real-Life Applications of Statistical Learning

Statistical learning methods can be applied to many real-world problems involving prediction, understanding relationships between variables, and discovering patterns in data. The following examples illustrate applications of classification, regression, and cluster analysis.

1.1 Classification Applications

Classification is a supervised learning method used when the response variable is categorical. The objective is to assign an observation to one of two or more predefined categories.

1.1.1 1. Email Spam Detection

Response: The response is categorical, with two possible outcomes: *spam* or *not spam*. Therefore, this is a binary classification problem.

Predictors: Possible predictors include the frequency of certain words, the presence of suspicious phrases, the sender's address, the number of links in the email, the use of capital letters, and whether the email contains HTML content or attachments.

Goal: Prediction

The primary goal is prediction. The objective is to correctly classify new incoming emails as either spam or legitimate. The main concern is how accurately the model performs on new, unseen emails rather than understanding why particular words or email characteristics are associated with spam.

1.1.2 2. Medical Disease Diagnosis

Response: The response could be *disease* or *no disease*, making this a binary classification problem.

Predictors: Potential predictors include the patient's age, blood pressure, cholesterol level, symptoms, medical test results, weight, and other relevant health measurements.

Goal: Inference

The goal can be inference when researchers or medical professionals are interested in understanding which factors are associated with the presence of a disease. For example, they may want to determine whether age, blood pressure, or cholesterol level is strongly related to the likelihood of having a particular disease. However, the same classification approach could also be used for prediction when diagnosing a new patient.

1.1.3 3. Telecom Customer Churn

Response: The response is categorical, with two possible outcomes: *churn* or *does not churn*.

Predictors: Potential predictors include data usage, number of customer complaints, subscription type, payment history, call frequency, customer tenure, and the frequency of service interruptions.

Goal: Prediction

The primary goal is prediction. A telecommunications company can use classification to identify customers who are likely to leave the service. The company can then take preventive measures, such as providing targeted offers or improving customer support, to reduce customer churn. The emphasis is therefore on accurately identifying future churners.

1.2 Regression Applications

Regression is a supervised learning method used when the response variable is quantitative. It is useful for understanding relationships between a numerical outcome and one or more predictors, as well as for predicting numerical outcomes.

1.2.1 1. Predicting House Prices

Response: The response is the selling price of a house, which is a quantitative and continuous variable.

Predictors: Possible predictors include the size of the house, number of bedrooms and bathrooms, lot size, neighborhood, age of the house, quality of nearby schools, and distance from the city center.

Goal: Prediction

The goal is prediction. For example, a real estate company may want to estimate the price of a house that is being put on the market. The main objective is to produce an accurate estimate of the price based on the characteristics of the house.

1.2.2 2. Employee Salary Analysis

Response: The response is the employee's salary, which is a quantitative variable.

Predictors: Potential predictors include years of work experience, education level, job position or level, age, location, and type of occupation.

Goal: Inference

The goal is inference when the organization wants to understand the relationship between employee characteristics and salary. For example, the organization may want to determine whether additional years of experience or higher education are associated with higher salaries and how strongly these factors are related to salary.

1.2.3 3. Advertising Expenditure and Sales

Response: The response is sales revenue, which is a quantitative variable.

Predictors: The predictors could include advertising expenditure on television, radio, newspapers, online advertising, and other marketing channels.

Goal: Inference

The goal is inference when a company wants to understand how its advertising expenditure is related to sales. For example, the company may want to determine which advertising channels have the strongest relationship with sales and how changes in advertising expenditure are associated with changes in sales. This information can help the company make informed decisions about how to allocate its advertising budget.

1.3 Cluster Analysis Applications

Cluster analysis is an unsupervised learning technique. Unlike classification and regression, it does not require a predefined response variable. Instead, the objective is to identify natural groups or clusters of observations that have similar characteristics.

1.3.1 1. Customer and Market Segmentation

Businesses and retailers can use cluster analysis to group customers according to similarities in their purchasing behavior, demographics, browsing habits, spending patterns, and purchase frequency.

For example, the analysis may identify groups such as *budget-conscious customers, luxury buyers, frequent customers,* and *occasional shoppers.* Businesses can then use these groups to develop targeted marketing campaigns, personalized recommendations, and appropriate pricing or promotional strategies.

1.3.2 2. Gene Expression Analysis

Cluster analysis can be used in biological and medical research to identify groups of genes or patients that exhibit similar patterns.

For example, researchers can group genes according to their expression levels across different biological conditions. Genes that behave similarly may have related biological functions or be regulated by similar mechanisms. Similarly, patients can be grouped according to their gene expression profiles, potentially helping researchers identify different disease subtypes.

1.3.3 3. Document and Topic Clustering

Cluster analysis can be used by search engines, news organizations, and information management systems to organize large collections of documents according to their content.

For example, news articles can be automatically grouped into topics such as *politics, sports, technology, business,* and *entertainment* based on similarities in the words and content they contain. This allows large amounts of information to be organized without requiring each document to be manually assigned to a predefined category.

2 K-Nearest Neighbors Classification

The table below provides a training data set containing six observations, three predictors, and one qualitative response variable. We wish to use K-nearest neighbors (KNN) to make a prediction for the response Y when $X_1 = X_2 = X_3 = 0$.

2.1 (a) Euclidean Distances to the Test Point

The Euclidean distance between an observation (X_1, X_2, X_3) and the test point $(0, 0, 0)$ is given by

$$d = \sqrt{(X_1 - 0)^2 + (X_2 - 0)^2 + (X_3 - 0)^2} = \sqrt{X_1^2 + X_2^2 + X_3^2}.$$

The distances for all six observations are shown in Table 1.

Table 1: Euclidean distances from each observation to the test point $(0, 0, 0)$.

Obs.	X_1	X_2	X_3	Y	**Distance**
1	0	3	0	Red	$\sqrt{0^2 + 3^2 + 0^2} = \mathbf{3.00}$
2	2	0	0	Red	$\sqrt{2^2 + 0^2 + 0^2} = \mathbf{2.00}$
3	0	1	3	Red	$\sqrt{0^2 + 1^2 + 3^2} = \mathbf{3.16}$
4	0	1	2	Green	$\sqrt{0^2 + 1^2 + 2^2} = \mathbf{2.24}$
5	-1	0	1	Green	$\sqrt{(-1)^2 + 0^2 + 1^2} = \mathbf{1.41}$
6	1	1	1	Red	$\sqrt{1^2 + 1^2 + 1^2} = \mathbf{1.73}$

Ranking the observations from closest to farthest gives

$$\boxed{\text{Obs. } 5 < \text{Obs. } 6 < \text{Obs. } 2 < \text{Obs. } 4 < \text{Obs. } 1 < \text{Obs. } 3}$$

Therefore, Observation 5 is the closest observation to the test point.

2.2 (b) Prediction with $K = 1$

The closest observation to the test point is Observation 5, with a distance of approximately 1.41. Observation 5 has the response class **Green**.

Therefore, with $K = 1$,

$$\boxed{Y = \text{Green}}$$

With $K = 1$, KNN assigns the test observation the class of its single nearest neighbor. Since Observation 5 is the closest observation and its class is Green, the predicted class is Green.

2.3 (c) Prediction with $K = 3$

The three closest observations are:

1. Observation 5: distance 1.41 $\rightarrow$ Green
2. Observation 6: distance 1.73 $\rightarrow$ Red
3. Observation 2: distance 2.00 $\rightarrow$ Red

Among these three nearest neighbors, two observations are Red and one observation is Green.

$$\text{Red} = 2, \qquad \text{Green} = 1$$

Therefore, the majority class is Red, and the prediction is

$$\boxed{Y = \text{Red}}$$

KNN with $K = 3$ assigns the test point to the class receiving the majority vote among its three nearest neighbors. Since two of the three nearest neighbors are Red, the predicted class is Red.

2.4 (d) Effect of a Highly Non-Linear Bayes Decision Boundary

If the Bayes decision boundary is highly non-linear, we would generally expect the best value of K to be **small**.

The value of K controls the flexibility of the KNN classifier. A small value of K considers only a small number of nearby observations, allowing the classifier to adapt closely to local patterns in the data. Therefore, it can produce a more flexible and complex decision boundary.

In contrast, a large value of K considers many observations when making each prediction. This produces a smoother decision boundary and can overlook important local patterns. If the true Bayes decision boundary is highly non-linear, excessive smoothing may result in underfitting and higher bias.

Therefore, because the Bayes decision boundary is highly non-linear, we would generally prefer a smaller value of K so that the KNN classifier has enough flexibility to capture the complex boundary.

However, choosing $K = 1$ is not automatically optimal. Very small values of K can make the model sensitive to noise and lead to high variance. In practice, the best value of K would typically be selected using cross-validation.

$$\boxed{\text{Highly non-linear boundary} \quad \Longrightarrow \quad \text{generally smaller } K}$$

3 Linear Regression versus Cubic Regression

Suppose that we collect a data set consisting of $n = 100$ observations, with a single predictor X and a quantitative response Y. We fit both a linear regression model and a cubic regression model of the form

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon.$$

3.1 (a) True Relationship is Linear — Training RSS

The cubic regression will have a training RSS that is **less than or equal to** the training RSS of the linear regression.

The reason is that the cubic model contains the linear model as a special case. The cubic model is

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon.$$

If we set

$$\beta_2 = 0 \quad \text{and} \quad \beta_3 = 0,$$

the cubic model becomes

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

which is exactly the linear regression model.

Therefore, the cubic model has all the flexibility of the linear model, plus the ability to model quadratic and cubic effects. When minimizing the training RSS, the cubic model can always choose $\beta_2 = 0$ and $\beta_3 = 0$ if that gives the best fit. Consequently, adding these additional predictors cannot increase the minimum training RSS.

Thus,

$$\boxed{\text{Training RSS}_{\text{cubic}} \leq \text{Training RSS}_{\text{linear}}}$$

The cubic model may have a strictly lower training RSS, or the two models may have exactly the same training RSS.

3.2 (b) True Relationship is Linear — Test RSS

There is **not enough information to determine which model will have the lower test RSS**.

Although the true relationship is linear, the cubic model is more flexible and can fit the training data more closely. However, this additional flexibility can cause the cubic model to fit random noise in the training data, resulting in higher variance and potentially poorer performance on new observations.

The linear model has the correct functional form in this case, so it generally has lower variance and does not need to estimate the additional quadratic and cubic terms.

However, test RSS depends on the **bias-variance tradeoff**. Therefore, without additional information about the data, noise level, and estimated coefficients, we cannot determine which model will have lower test RSS.

Thus,

$$\boxed{\text{Not enough information to tell}}$$

The cubic model could have a lower, equal, or higher test RSS than the linear model.

3.3 (c) True Relationship is Not Linear — Training RSS

The cubic regression will have a training RSS that is **less than or equal to** the training RSS of the linear regression.

This is true regardless of whether the true relationship is linear or nonlinear. The cubic model contains the linear model as a special case because setting

$$\beta_2 = 0 \quad \text{and} \quad \beta_3 = 0$$

reduces the cubic model to the linear model.

Therefore, when minimizing the training RSS, the cubic model has at least as many possible solutions as the linear model. It can reproduce the linear model if necessary and can additionally use the X^2 and X^3 terms to capture nonlinear patterns in the training data.

Thus,

$$\boxed{\text{Training RSS}_{\text{cubic}} \leq \text{Training RSS}_{\text{linear}}}$$

The cubic model will generally have a lower training RSS when the nonlinear relationship can be captured by the cubic terms, although the two RSS values could theoretically be the same.

3.4 (d) True Relationship is Not Linear — Test RSS

There is **not enough information to determine which model will have the lower test RSS**.

Because the true relationship is nonlinear, the cubic model has an advantage because it can capture certain types of nonlinear relationships that the linear model cannot. This can result in lower bias.

However, we are not told **how far the true relationship is from linear**.

If the true relationship is only slightly nonlinear, the simpler linear model may perform as well as or better than the cubic model on test data because the cubic model has greater variance and may fit noise unnecessarily.

On the other hand, if the true relationship is strongly nonlinear and can be well approximated by a cubic function, the cubic model may have substantially lower test RSS because it can capture the underlying nonlinear pattern.

Therefore, without knowing the exact form of the nonlinear relationship, the amount of noise, and the resulting bias-variance tradeoff, we cannot determine which model will have lower test RSS.

Thus,

$$\boxed{\text{Not enough information to tell}}$$